ECON2103 Microeconomics

Chapter 5 Uncertainty and Consumer Behavior

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Outline

1. Describing Risk
2. Preferences Toward Risk
3. Reducing Risk
4. Summary and Exercises

Probability

probability - Likelihood that a given outcome will occur.

- One *objective* interpretation of probability relies on the frequency with which certain events tend to occur.
- *Subjective probability* is the perception that an outcome will occur.

Two important measures that help us describe and compare risky choices are *expected value* and *variability* of the possible outcomes.

Expected Value

expected value - Probability-weighted average of the payoffs associated with possible outcomes.

- **payoff** - Value associated with a possible outcome.
- The expected value measures the *central tendency* - the payoff or value that we would expect on average.

When there are n possible outcomes, the expected value is

$$E(X) = Pr_1 X_1 + Pr_2 X_2 + \cdots + Pr_n X_n$$

Variability

variability - Extent to which possible outcomes of an uncertain event differ.

TABLE 5.1 INCOME FROM SALES JOBS

	OUTCOME 1		OUTCOME 2		
	PROBABILITY	INCOME ($)	PROBABILITY	INCOME ($)	EXPECTED INCOME ($)
Job 1: Commission	.5	2000	.5	1000	1500
Job 2: Fixed Salary	.99	1510	.01	510	1500

deviation - Difference between expected payoff and actual payoff.

TABLE 5.2 DEVIATIONS FROM EXPECTED INCOME ($)

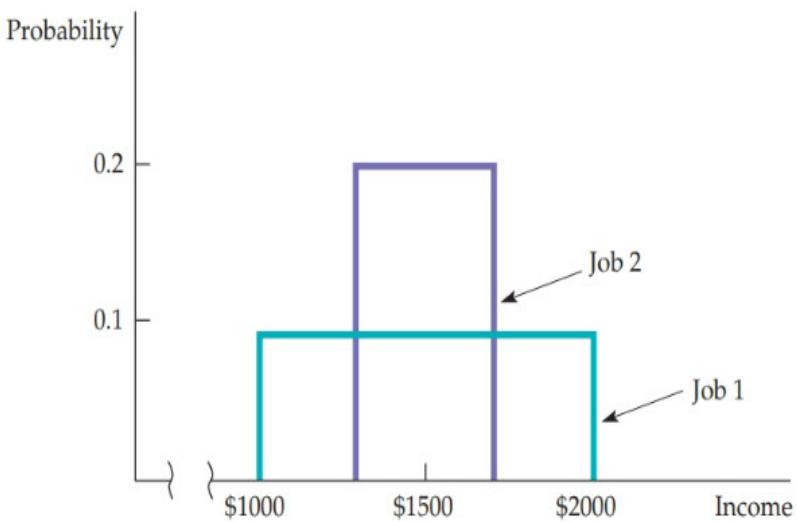
	OUTCOME 1	DEVIATION	OUTCOME 2	DEVIATION
Job 1	2000	500	1000	−500
Job 2	1510	10	510	−990

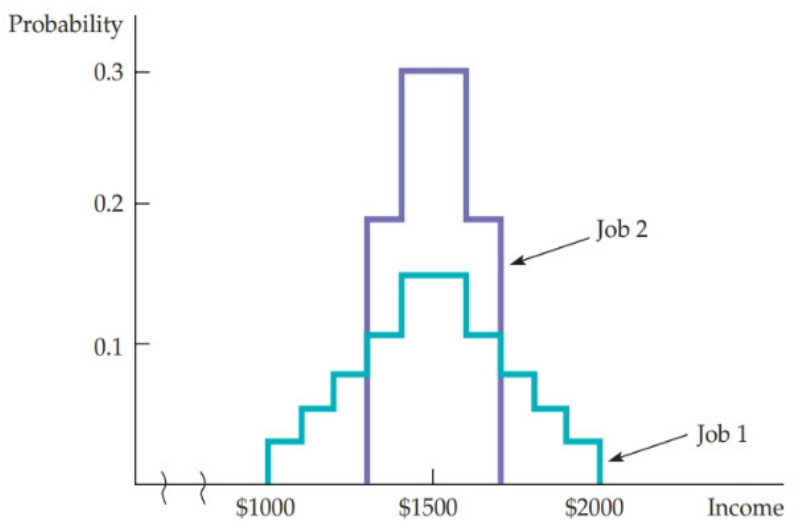
Standard Deviation

standard deviation - Square root of the weighted average of the squares of the deviations of the payoffs associated with each outcome from their expected values

TABLE 5.3 CALCULATING VARIANCE ($)

	OUTCOME 1	DEVIATION SQUARED	OUTCOME 2	DEVIATION SQUARED	WEIGHTED AVERAGE DEVIATION SQUARED	STANDARD DEVIATION
Job 1	2000	250,000	1000	250,000	250,000	500
Job 2	1510	100	510	980,100	9900	99.50





Decision Making

TABLE 5.4	INCOMES FROM SALES JOBS—MODIFIED ($)					
	OUTCOME 1	DEVIATION SQUARED	OUTCOME 2	DEVIATION SQUARED	EXPECTED INCOME	STANDARD DEVIATION
Job 1	2100	250,000	1100	250,000	1600	500
Job 2	1510	100	510	980,100	1500	99.50

Job 1: Expected Income = \$1600; Standard Deviation = \$500
Job 2: Expected Income = \$1500; Standard Deviation = \$99.5

An aggressive entrepreneur who doesn't mind taking risks might choose Job 1, while a more conservative person might choose Job 2.

Expected Utility

We measure payoffs in terms of utility rather than dollars.

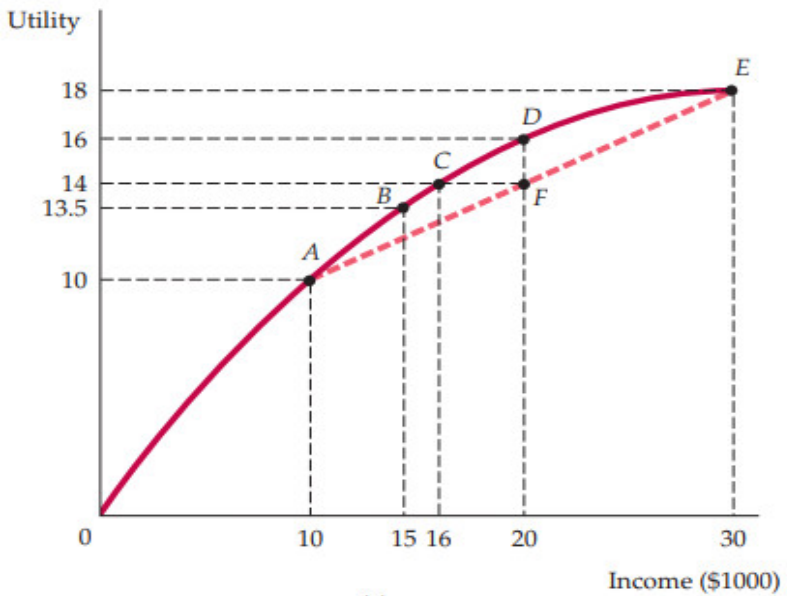
expected utility - Sum of the utilities associated with all possible outcomes, weighted by the probability that each outcome will occur.

Risk Averse

risk averse - Condition of preferring a certain income to a risky income with the same expected value. 风险规避者

- Marginal utility diminishes as income increases.
- For a risk-averse person, losses are more important than gains.

损失更重要：损失对效用的负面影响大于同等收益的正面影响。



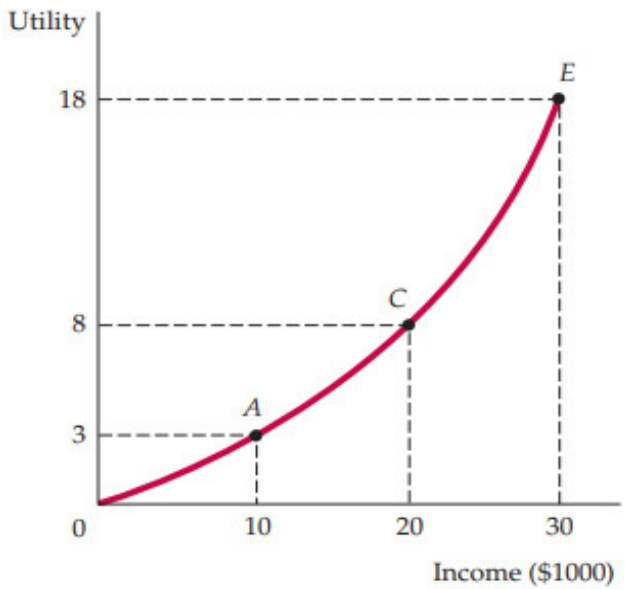
图中左侧的效用函数是凹形的，显示了边际效用递减。例如，点 D 表示确定收入带来的效用，而点 F 表示具有相同期望收入的风险选择。由于 U(D) > U(F)，消费者选择 D。

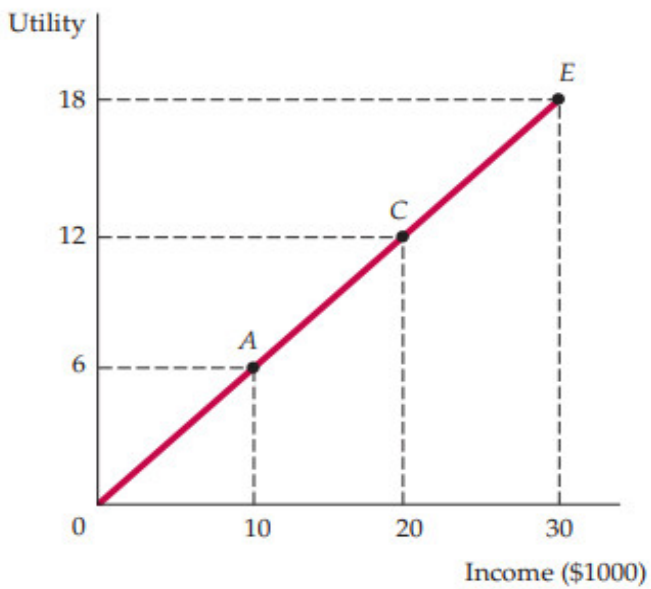
The consumer is risk averse because she would prefer D to F.

Risk Loving

risk loving - Condition of preferring a risky income to a certain income with the same expected value.

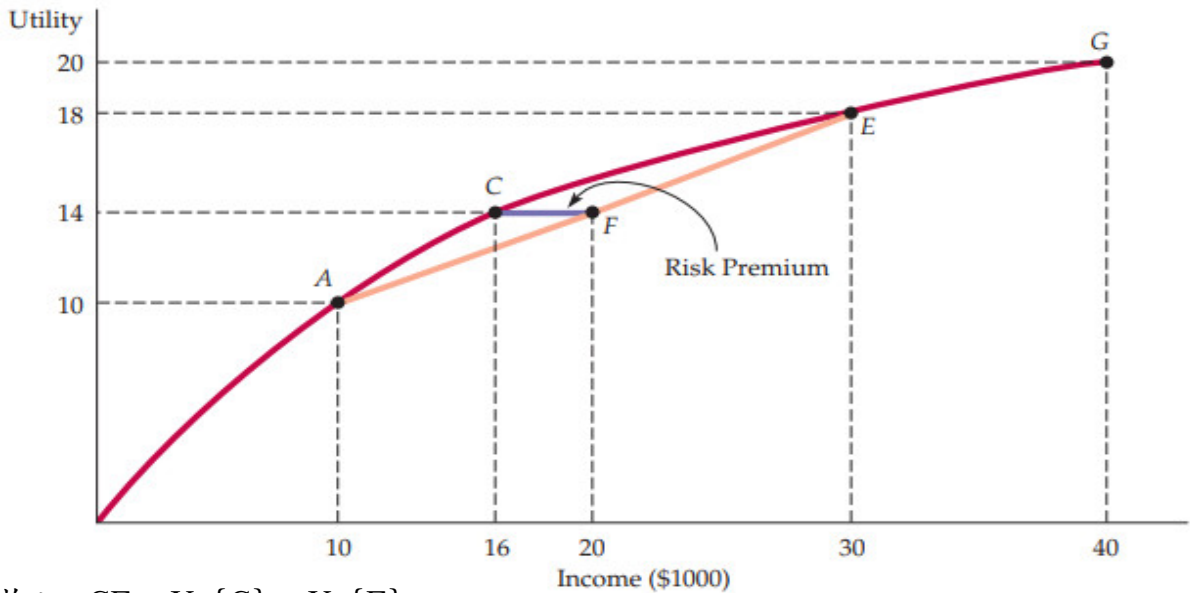
risk neutral - Condition of being indifferent between a certain income and an uncertain income with the same expected value.





Risk Premium

risk premium - Maximum amount of money that a risk-averse person will pay to avoid taking a risk.



确定效用的收入 CF = Y_{C} - Y_{F}。
它测量了一个人愿意为了避免风险而放弃的收入。

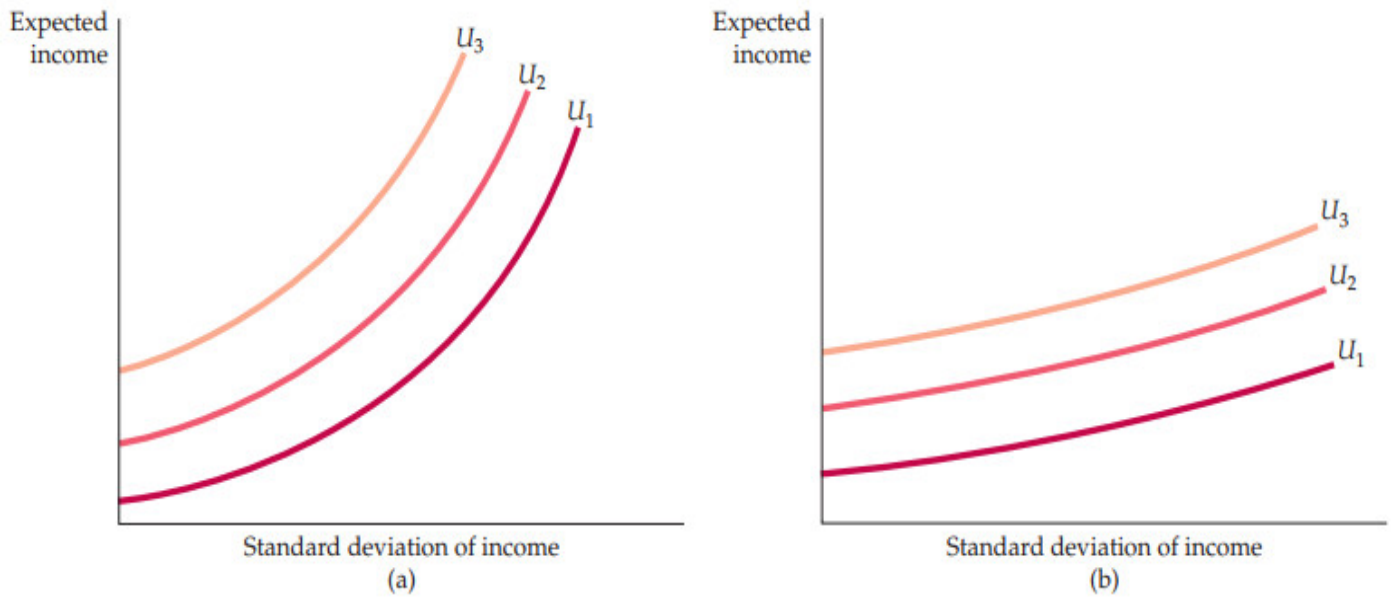
The risk premium, *CF*, measures the amount of expected income that an individual would give up to leave her indifferent between a risky choice and a certain one.

Risk Aversion and Income

- The extent of an individual's risk aversion depends on the nature of the risk and on the person's income. Other things being equal, risk-averse people prefer a smaller variability of outcomes.
- The greater the variability of income, the more the person would be willing to pay to avoid the risky situation.
  - Consider a second risky job: .5 probability of receiving an income of $40,000, with a utility level of 20, and a .5 probability of getting an income of $0, with a utility level of 0.

收入越高，边际效用递减，风险规避程度可能会减弱。
收入变异性越大，风险规避者越倾向于支付更多费用避免风险。

Risk Aversion and Indifference Curves



Which plot applies to a more risk-averse person?

1.左图 (a): 更风险规避的人
无差异曲线更陡峭，说明对于相同期望收入，更大的标准差（风险）需要更高的补偿。
2.右图 (b): 风险偏好较低或风险中立。
无差异曲线较平缓，说明对于更高的风险，愿意接受的补偿较低。

Diversification

diversification - Practice of reducing risk by allocating resources to a variety of activities whose outcomes are not closely related.

TABLE 5.5	INCOME FROM SALES OF APPLIANCES ($)	
	HOT WEATHER	COLD WEATHER
Air conditioner sales	30,000	12,000
Heater sales	12,000	30,000

If you sell only air conditioners or only heaters, your expected income will be _____.

If you diversify by dividing your time evenly between the two products, you income will be _____, regardless of the weather.

negatively correlated variables - Variables having a tendency to move in opposite directions.

The Stock Market

mutual fund - Organization that pools funds of individual investors to buy a large number of different stocks or other financial assets.

positively correlated variables - Variables having a tendency to move in the same direction.

Insurance

- Risk-averse people are willing to pay to avoid risk.
- If the cost of insurance is equal to the expected loss, risk-averse people will buy enough insurance to recover fully from any financial losses they might suffer.

TABLE 5.6	THE DECISION TO INSURE ($)			
INSURANCE	BURGLARY (PR = .1)	NO BURGLARY (PR = .9)	EXPECTED WEALTH	STANDARD DEVIATION
No	40,000	50,000	49,000	3000
Yes	49,000	49,000	49,000	0

For a risk-averse individual, losses count more (in terms of changes in utility) than gains. A risk-averse homeowner, therefore, will enjoy higher utility by purchasing insurance.

Law of Large Numbers and Actuarial Fairness

- Insurance companies are firms that offer insurance because they know that when they sell a large number of policies, they face relatively little risk.
  - Although single events may be random and largely unpredictable, the average outcome of many similar events can be predicted.
- **actuarially fair** - Characterizing a situation in which an insurance premium is equal to the expected payout.
  - If there are a sufficient number of insurance companies to make the market competitive, these premiums will be close to actuarially fair level.

The Value of Information

value of complete information - Difference between the expected value of a choice when there is complete information and the expected value when information is incomplete.

Example: \text{Value of Complete Information} = \text{Expected Value (Complete Info)} - \text{Expected Value (Uncertainty)}

- If you order 100 suits, your cost is $180 per suit. If you order 50 suits, your cost increases to $200.
- You will be selling suits for $300 each.
- All suits not sold can be returned, but for only half of what you paid for them.
- There is a .5 probability that you will sell 100 suits and a .5 probability that you will sell 50.

The Value of Information (Con't)

TABLE 5.7	PROFITS FROM SALES OF SUITS ($)		
	SALES OF 50	SALES OF 100	EXPECTED PROFIT
Buy 50 suits	5000	5000	5000
Buy 100 suits	1500	12,000	6750

Expected value with complete information:

Less: Expected value with uncertainty (buy 100 suits):

Equals: Value of complete information:

Summary

- probability, expected value, variability, standard deviation
- risk averse, risk loving, risk neutral
- Reducing risk
  - diversification
  - insurance
  - obtaining more information

Exercises

1. Which of the following is NOT a generally accepted measure of the riskiness of an investment?
   a. Standard deviation
   b. Expected value
   c. Variance
   d. none of the above

Exercises (Con't)

2. Other things equal, expected income can be used as a direct measure of well-being:
   a. always.
   b. no matter what a person's preference to risk.
   c. if and only if individuals are not risk-loving.
   d. if and only if individuals are risk averse.
   e. if and only if individuals are risk neutral.

无论个人的风险偏好如何，期望收入都可以作为福利的直接衡量标准。

Exercises (Con't)

3. John Brown's utility of income function is $U = \log(I + 1)$, where I represents income. From this information you can say that
   a. John Brown is risk neutral.
   b. John Brown is risk loving.
   c. John Brown is risk averse.
   d. We need more information before we can determine John Brown's preference for risk.

Exercises (Con't)

4. Smith just bought a house for $250,000. Earthquake insurance, which would pay $250,000 in the event of a major earthquake, is available for $25,000. Smith estimates that the probability of a major earthquake in the coming year is 10 percent, and that in the event of such a quake, the property would be worth nothing. The utility (U) that Smith gets from income (I) is $U(I) = I^{0.5}$. Should Smith buy the insurance?
   a. Yes
   b. No
   c. Smith is indifferent.
   d. We need more information on Smith's attitude toward risk.

Exercises (Con't)

5. We may not be able to fully remove risk by diversification if
   a. a completely risk-free asset does not exist.
   b. the asset returns in our portfolio are positively correlated.
   c. buying stock on margin is not allowed by financial regulators.
   d. none of the above.

a. 完全无风险资产不存在。
b. 投资组合中的资产收益是正相关的。
c. 金融监管机构不允许买空股票。

分散化的关键在于投资于非相关或负相关的资产，以抵消各资产的波动风险。
如果资产收益是正相关的，那么它们的波动方向一致，分散化无法显著降低风险。
选项 a 和 c 与风险分散关系不大，因此正确答案是 b。